

Multimodal Analysis of Social Media for Disaster Response and Management

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PROBLEM STATEMENT

Social media often becomes the last space to be heard for many disaster victims. However, countless posts—blurry images of disaster urgent messages, appeals for rescue are lost within the countless of posts online.

The key question then will be: "How do we create a model to detect not only whether a post is disaster-related, but also the severity of the situation?" This challenge requires a multimodal framework that can read both textual data and understand the visual images for each post. Jointly analyzing text and images should enable the system to identify real disaster-related content, classify its severity, and relay situational information far earlier than traditional field reports to help responders act more quickly.

OBJECTIVES

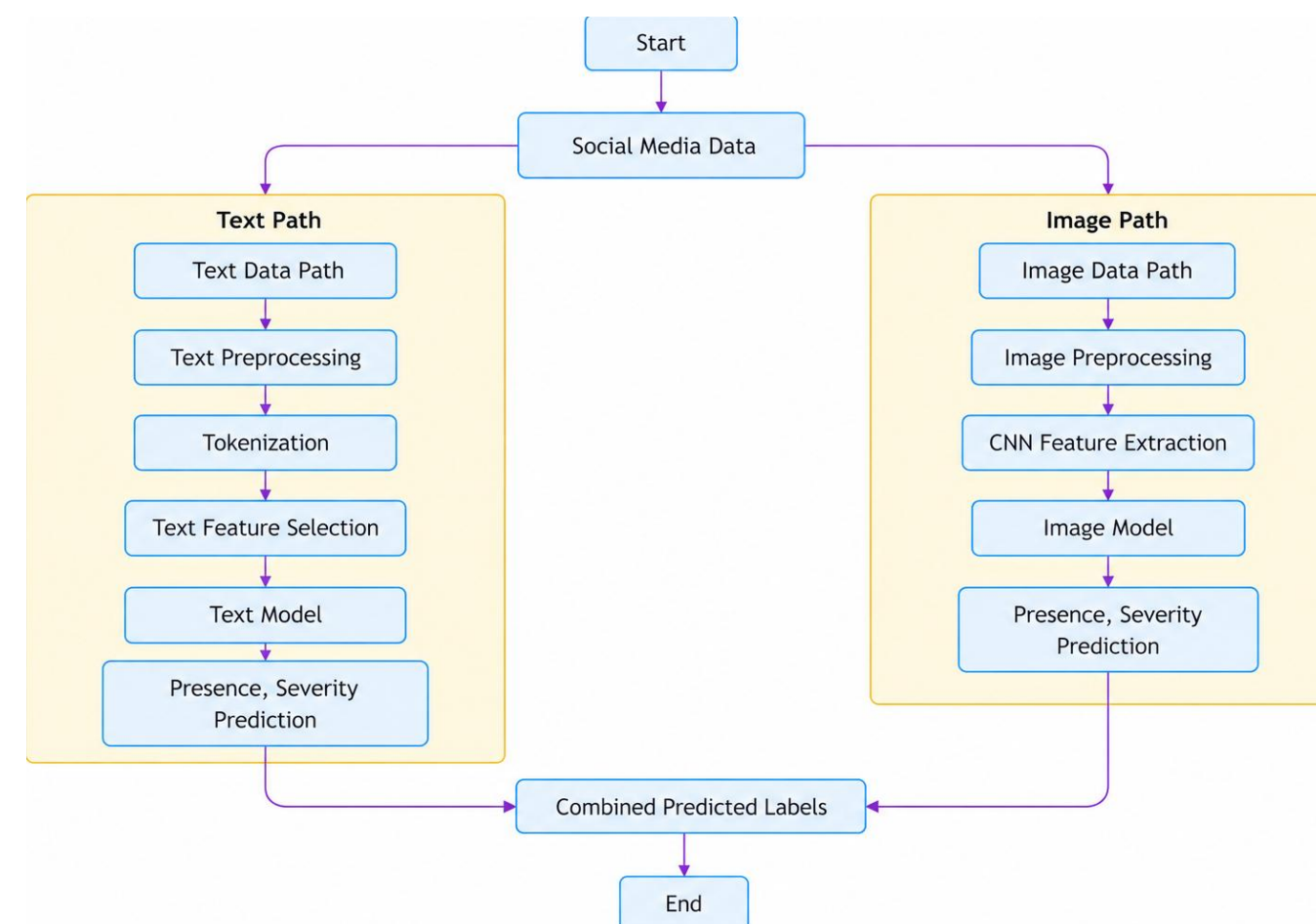
- To build a system that can actually "read" social media posts—both the text data and the image data—to detect disasters as they occur in real time.
- Classify the disaster related social media data into the severity level (high , medium and low) using state of the art models.
- Filter out noisy or misinformed and unrelated data that may not contribute to the overall model building.

METHODOLOGY

The proposed system uses a multimodal deep learning approach for disaster detection and severity classification using both text and images collected from social media platforms such as Twitter and Reddit. The textual data is preprocessed through cleaning, tokenization, and normalization before being passed to a BERT-based model for relevance and severity classification. Simultaneously, image data is resized, normalized, and processed using CNN and EfficientNet-based models to identify disaster categories such as flood, fire, and earthquake.

The extracted text and image features are then analyzed to classify posts into disaster-related and non-disaster categories while also estimating severity levels such as low, medium, and high impact. The models are trained and evaluated using metrics including accuracy, precision, recall, F1-score, confusion matrix, and ROC curve to ensure reliable real-world performance.

WORKFLOW:

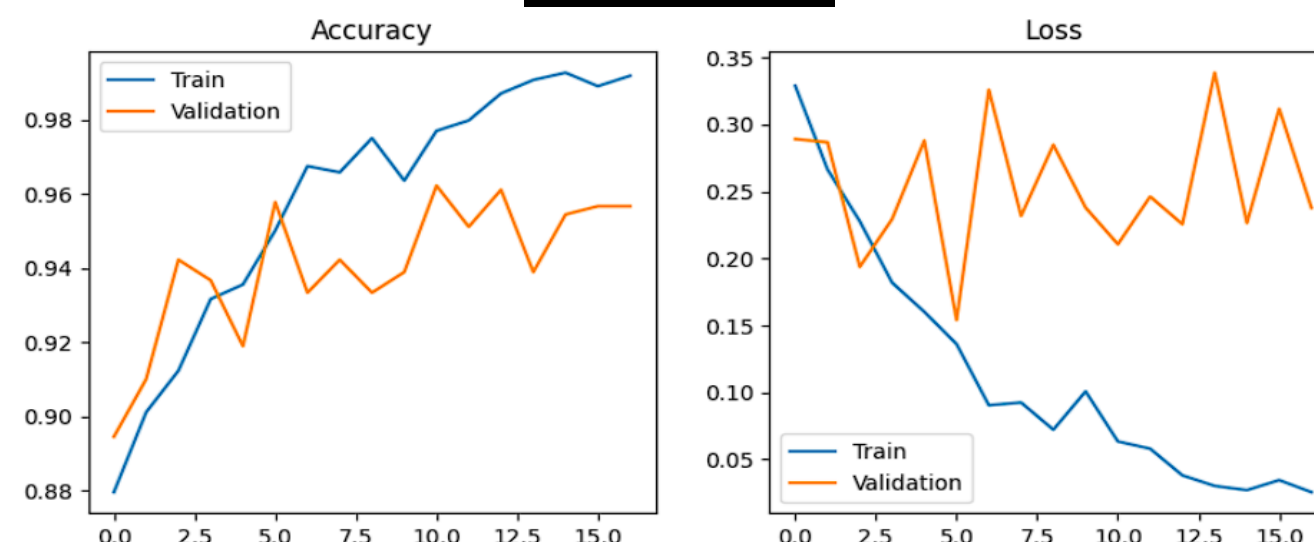


IMPLEMENTATION DETAIL

The proposed multimodal disaster detection system was implemented using NLP and Deep Learning techniques for analyzing both text and images from social media data. Text data was preprocessed and classified using a BERT-based model, while disaster images were processed using CNN and EfficientNet architectures for disaster classification.

The models were trained and evaluated using metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. The implementation was carried out in Python using Google Colab with TensorFlow, Keras, OpenCV, Scikit-learn, and Hugging Face Transformers.

RESULTS



The proposed multimodal disaster detection system effectively identified disaster-related content and classified severity levels using both text and images. The text model successfully filtered relevant posts and categorized severity into low, medium, and high impact classes.

The CNN and EfficientNet image models showed strong performance in detecting disasters such as flood, fire, and earthquake. Evaluation metrics including accuracy, precision, recall, F1-score, confusion matrix, and ROC curve indicated reliable and consistent classification performance.

CONCLUSION & FUTURE SCOPE

CONCLUSION:

- Developed a multimodal disaster detection system using both text and image analysis.
- Integrated BERT-based NLP with CNN and EfficientNet models for disaster classification.
- Successfully detected disasters such as flood, fire, and earthquake with reliable performance.
- Improved disaster classification accuracy and situational awareness for faster emergency response and disaster management.

FUTURE SCOPE and ENHANCEMENTS:

- Advanced multimodal fusion and cross-modal attention techniques can be used to improve the connection between text and image data. This can increase disaster detection accuracy and contextual understanding.
- The system can be extended for real-time disaster monitoring using live social media streams and automated alerts. Future improvements may also include Explainable AI and mobile/edge deployment for faster response.

REFERENCES

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